Sri Lanka Institute of Information Technology



IT3091 – Machine Learning

Year 3, Semester 1 - 2026

Data Dictionary

EduPredict: Student Performance Prediction
Guided Data Track – Education (Student Performance)

Item	Details
Group	2026-AI-33
Campus	Malabe
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Table P.1: Document submission metadata

Group Members

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Table P.2: Group 2026-AI-33 member details

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1. Introduction

1.1 Scope of the dictionary

This dictionary defines the 33 raw columns in data/student-mat.csv and their role in the implemented project. It separates source descriptions from observed Mathematics-file values, records preprocessing choices and identifies unresolved interpretation issues. One row represents one student; the 395-row file has no missing cells or exact duplicate full rows.

1.2 Model inputs and target

Main model inputs: 30 columns excluding G1, G2 and G3. Later-stage inputs: 32 columns excluding G3. G1/G2 are available only after their respective grading periods. G3 is the target and is never an input. The stored target is integer-valued; predictions are continuous.

2. Variable Definitions

2.1 Student and family information

Variable	Type	Role	Meaning and codes	Observed
school	Binary nominal	Both	School. GP = Gabriel Pereira; MS = Mousinho da Silveira	GP, MS
sex	Binary nominal	Both	Recorded sex. F = female; M = male	F, M
age	Discrete numeric	Both	Age in years. Documented range 15–22	15–22
address	Binary nominal	Both	Home area. U = urban; R = rural	R, U
famsize	Binary nominal	Both	Family size. LE3 = at most 3; GT3 = more than 3	GT3, LE3
Pstatus	Binary nominal	Both	Parental cohabitation. T = together; A = apart	A, T
Medu	Ordinal code	Both	Mother education. 0 none; 1 primary grade 4; 2 grades 5–9; 3 secondary; 4 higher	0–4
Fedu	Ordinal code	Both	Father education. Same education codes as Medu	0–4
Mjob	Nominal	Both	Mother occupation. teacher; health; services; at_home; other	at_home, health, other, services, teacher
Fjob	Nominal	Both	Father occupation. Same occupation codes as Mjob	at_home, health, other, services, teacher
reason	Nominal	Both	School choice reason. home = proximity; reputation; course; other	course, home, other, reputation
guardian	Nominal	Both	Guardian. mother; father; other	father, mother, other

Table 2.1: Student and family information

2.2 Study history and support

Variable	Type	Role	Meaning and codes	Observed
traveltime	Ordinal code	Both	Travel time. 1 <15 min; 2 15–30 min; 3 30–60 min; 4 >60 min	1–4
studytime	Ordinal code	Both	Weekly study time. 1 <2 h; 2 2–5 h; 3 5–10 h; 4 >10 h	1–4
failures	Discrete code	Both	Past class failures. Observed codes 0,1,2,3; source coding conflict remains unresolved	0–3
schoolsup	Binary nominal	Both	Extra school support. yes / no	no, yes
famsup	Binary nominal	Both	Family learning support. yes / no	no, yes
paid	Binary nominal	Both	Paid subject classes. yes / no	no, yes
activities	Binary nominal	Both	Extracurricular participation. yes / no	no, yes
nursery	Binary nominal	Both	Nursery attendance. yes / no	no, yes
higher	Binary nominal	Both	Higher education aspiration. yes / no	no, yes
internet	Binary nominal	Both	Home internet. yes / no	no, yes
romantic	Binary nominal	Both	Romantic relationship. yes / no	no, yes

Table 2.2: Study history and support

2.3 Behaviour, attendance and grades

Variable	Type	Role	Meaning and codes	Observed
famrel	Ordinal code	Both	Family relationship quality. 1 very bad through 5 excellent	1–5
freetime	Ordinal code	Both	Free time after school. 1 very low through 5 very high	1–5
goout	Ordinal code	Both	Going out with friends. 1 very low through 5 very high	1–5
Dalc	Ordinal code	Both	Workday alcohol use. 1 very low through 5 very high	1–5
Walc	Ordinal code	Both	Weekend alcohol use. 1 very low through 5 very high	1–5
health	Ordinal code	Both	Current health. 1 very bad through 5 very good	1–5
absences	Discrete numeric	Both	School absences. Documented range 0–93; collection window unconfirmed	0–75
G1	Discrete numeric	Later only	First period grade. Documented scale 0–20; later-stage input only	3–19
G2	Discrete numeric	Later only	Second period grade. Documented scale 0–20; later-stage input only	0–19
G3	Discrete numeric	Target	Final grade. Documented scale 0–20; target only; recorded zeros retained	0–20

Table 2.3: Behaviour, attendance and grades

3. Preprocessing Used by the Models

Feature group	Implementation
Binary nominal 13	school GP=0/MS=1; sex F=0/M=1; address R=0/U=1; famsize LE3=0/GT3=1; Pstatus A=0/T=1. Eight yes/no features map no=0 and yes=1.
Nominal 4	Mjob, Fjob, reason and guardian use one-hot encoding. Model notebooks set drop=first and handle_unknown=ignore; an unseen category becomes all zeros in its group.
Ordinal 10	Medu, Fedu, traveltime, studytime, famrel, freetime, goout, Dalc, Walc and health retain ordered numerical codes; scaling depends on the model.
Other numerical inputs	age, failures and absences; G1/G2 are added explicitly only for the later-stage scenario.
Absence option	Search original absences versus absences_log = log1p(absences). The transformed feature replaces absences; it is not a new raw CSV column.
Scaling	StandardScaler for numerical and ordered inputs in Linear Regression/Ridge; no scaling for the tree models. Fit inside each training fold.

Table 3.1: Preprocessing by feature group

4. Interpretation, Data Quality and Sources

4.1 Interpretation and data quality notes

The observed Mathematics ranges differ from some source ranges: absences reaches 75 rather than the documented 93; G1 spans 3–19 and G2 spans 0–19, while the documented grade scale is 0–20. Observed values describe this sample, not admissible future limits.

The source description of failures gives a coding rule inconsistent with the CSV's 0,1,2,3 values. The project retains the observed codes and treats them numerically. The meaning of the highest code is not confirmed, so the dictionary does not assert that it means three or more failures. This ambiguity should be resolved before interpreting an effect or deploying a model.

The dataset contains 38 G3=0 records. Their causes are unknown; retain them rather than assume absence, dropout, missingness or data error. Rating scales are ordered codes, not precise physical measurements. Equal numerical spacing is a modelling approximation.

Input availability must be checked at the chosen prediction date, especially the absence accumulation window. Sensitive personal and family information requires appropriate access controls and an operational necessity review.

4.2 Definitions and implementation sources

Definitions: data/student.txt and the existing project dictionary. Observed values: data/student-mat.csv audited on 12 October 2026. Implementation: src/preprocessing.py and notebooks/03a through 03e. Dataset attribution: Cortez, P. (2008), Student Performance, UCI Machine Learning Repository, https://doi.org/10.24432/C5TG7T.